

Finite-Level S -Adèle Models, Finite BC-Type Systems, and Truncated Explicit Formulae

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Abstract

We study finite-level truncations of the adelic and Bost–Connes frameworks. For a finite set of places S containing ∞ , we introduce the truncated S -adèle ring

$$\mathbb{A}_{\mathbb{Q}}^{(S)} = \left(\prod_{v \in S} \mathbb{Q}_v \right) \times \left(\prod_{v \notin S} \mathbb{Z}_v \right)$$

and the associated quotient

$$X_S = \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{(S)}.$$

This space provides a natural arithmetic configuration space for finite Euler products and truncated explicit-formula identities. We prove a finite Euler product expansion and an exact finite von Mangoldt identity indexed by the semigroup generated by the primes in S .

We then propose, at a heuristic level, a separate finite Bost–Connes type dynamical space Y_S , built from the semigroup generated by the primes in S , whose partition function reproduces the same finite Euler product. In this way, the arithmetic space X_S and the BC-type dynamical space Y_S play complementary roles: the former controls the explicit formula, while the latter is the natural candidate for a genuine operator-algebraic realization.

The paper is therefore divided into two layers: a rigorous finite- S arithmetic core, and a heuristic/operator-algebraic program suggesting how these finite models may fit into a computable approximation scheme for Hilbert–Pólya type ideas.

Keywords: Bost–Connes system; adèle class space; explicit formula; finite Euler product; von Mangoldt function; Riemann hypothesis.

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1 Introduction

The Riemann hypothesis asserts that the nontrivial zeros of the Riemann zeta function lie on the critical line $\operatorname{Re}(s) = 1/2$. A longstanding idea, often associated with Hilbert and Pólya, is that these zeros should arise spectrally from a self-adjoint operator. The work of Bost and Connes, and later Connes, showed that noncommutative dynamical systems built from arithmetic data can encode zeta functions, Euler products, KMS states, and explicit-formula phenomena.

At the same time, the full adelic and operator-algebraic frameworks are technically delicate. The full adèle ring is an infinite restricted product, and the standard BC system is not literally formulated on the adèle class space $\mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}$. This suggests a more modest approach: truncate to finitely many primes and separate the arithmetic and dynamical roles cleanly.

The purpose of this note is to do exactly that.

We work with two finite- S objects:

- the arithmetic configuration space

$$X_S := \mathbb{Q}^\times \backslash \mathbb{A}_\mathbb{Q}^{(S)},$$

where

$$\mathbb{A}_\mathbb{Q}^{(S)} = \left(\prod_{v \in S} \mathbb{Q}_v \right) \times \left(\prod_{v \notin S} \mathbb{Z}_v \right),$$

and

- a finite BC-type dynamical space Y_S , introduced heuristically, based on the monoid generated by the primes in S .

The first object X_S is the natural finite adelic shadow of the adèle class space and is well suited to explicit-formula considerations. The second object Y_S is the natural finite dynamical stage for BC-type thermodynamics and partition functions. The main point of the paper is that these two finite- S structures produce the same Euler factors and the same monoid of S -supported integers, but in different mathematical languages.

Status of results. Sections 2–4 contain rigorous finite-level constructions and identities. Section 5 introduces the finite BC-type space Y_S and explains how it should reproduce the same arithmetic data; this section is heuristic and programmatic rather than theorem-level. Section 6 discusses the relation between X_S and Y_S , and Section 7 explains how the finite models might fit into a limiting Hilbert–Pólya style picture. Appendix A contains numerical experiments.

2 S -Adèles and the Arithmetic Configuration Space

Let S be a finite set of places of $\mathbb{Q}$ containing ∞ . Define

$$\mathbb{A}_\mathbb{Q}^{(S)} := \left(\prod_{v \in S} \mathbb{Q}_v \right) \times \left(\prod_{v \notin S} \mathbb{Z}_v \right).$$

Thus for places in S we retain the full local field, while outside S we keep only the maximal compact subring. Since S is finite, $\mathbb{A}_\mathbb{Q}^{(S)}$ is a locally compact ring. It is naturally an open subring of the full adèle ring $\mathbb{A}_\mathbb{Q}$.

Definition 2.1. The finite- S arithmetic configuration space is

$$X_S := \mathbb{Q}^\times \backslash \mathbb{A}_\mathbb{Q}^{(S)}.$$

This should be viewed as a finite-level truncation of the adèle class space.

Remark 2.2. The role of X_S in this paper is arithmetic rather than operator-algebraic. We do not build a crossed-product C^* -algebra directly from X_S . Instead, X_S serves as the natural finite adelic space on which the finite Euler products and truncated explicit formulas are read.

As S increases, the union

$$\bigcup_S \mathbb{A}_\mathbb{Q}^{(S)}$$

is dense in $\mathbb{A}_\mathbb{Q}$, and correspondingly the spaces X_S should be regarded as finite approximations to the full adèle class space.

3 Finite Euler Products and the Monoid Λ_S

Let S now denote a finite set of finite primes. Define the free commutative monoid

$$\Lambda_S := \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}.$$

Equivalently, Λ_S is the set of positive integers whose prime factors all lie in S . For $\beta > 1$, define the finite Euler product

$$Z_S(\beta) := \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

Proposition 3.1 (Finite Euler product expansion). *For $\beta > 1$,*

$$Z_S(\beta) = \sum_{n \in \Lambda_S} n^{-\beta}.$$

Proof. For each $p \in S$,

$$(1 - p^{-\beta})^{-1} = \sum_{k=0}^{\infty} p^{-k\beta},$$

which converges absolutely for $\beta > 1$. Since the product over S is finite, we may multiply the geometric series termwise:

$$\prod_{p \in S} \left(\sum_{k=0}^{\infty} p^{-k\beta} \right) = \sum_{(k_p)_{p \in S} \in \mathbb{N}^S} \prod_{p \in S} p^{-k_p \beta}.$$

Each multi-index (k_p) corresponds uniquely to an integer

$$n = \prod_{p \in S} p^{k_p} \in \Lambda_S,$$

so the right-hand side is

$$\sum_{n \in \Lambda_S} n^{-\beta}.$$

□

The logarithmic derivative of this finite Euler product gives the finite von Mangoldt expansion.

Corollary 3.2 (Finite logarithmic derivative identity). *For $\beta > 1$,*

$$-\frac{d}{d\beta} \log Z_S(\beta) = \sum_{n \in \Lambda_S} \frac{\Lambda(n)}{n^\beta},$$

where Λ is the von Mangoldt function.

Proof. We have

$$\log Z_S(\beta) = - \sum_{p \in S} \log(1 - p^{-\beta}).$$

Differentiating termwise gives

$$-\frac{d}{d\beta} \log Z_S(\beta) = \sum_{p \in S} \frac{(\log p) p^{-\beta}}{1 - p^{-\beta}} = \sum_{p \in S} \sum_{k \geq 1} (\log p) p^{-k\beta}.$$

Since $\Lambda(n) = \log p$ exactly when $n = p^k$, this is

$$\sum_{n \in \Lambda_S} \frac{\Lambda(n)}{n^\beta}.$$

□

Remark 3.3. Everything in this section is completely finite and completely rigorous. No operator algebra, KMS theory, or adelic analysis is needed here. This finite semigroup identity is the arithmetic backbone of the rest of the paper.

4 Finite Explicit Formula in Arithmetic Form

Let h be a Schwartz function on $\mathbb{R}$.

Theorem 4.1 (Finite explicit formula, arithmetic form). *Let S be a finite set of primes. Then*

$$\sum_{n \in \Lambda_S} \frac{\Lambda(n)}{\sqrt{n}} \left(h(\log n) + h(-\log n) \right) = \sum_{p \in S} \sum_{k \geq 1} \frac{\log p}{p^{k/2}} \left(h(k \log p) + h(-k \log p) \right).$$

Proof. Every element $n \in \Lambda_S$ factors uniquely as

$$n = \prod_{p \in S} p^{k_p}.$$

The von Mangoldt function satisfies

$$\Lambda(n) = \begin{cases} \log p, & \text{if } n = p^k, \\ 0, & \text{otherwise.} \end{cases}$$

Hence only prime powers contribute to the sum on the left. Reindexing by the contributing prime powers gives exactly the right-hand side. □

Remark 4.2. This theorem is the rigorous finite- S arithmetic core of the later trace-formula discussion. In the present paper it should be read as a finite exact identity, not yet as a theorem about traces of operators.

5 A Finite BC-Type Dynamical Space

We now introduce the dynamical object that should play the BC-type role at finite level.

5.1 The finite-prime monoid

Let

$$\mathbb{N}_S := \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}.$$

Thus $\mathbb{N}_S = \Lambda_S$ as a multiplicative monoid.

A natural finite-prime BC-type configuration space is

$$Y_S := \prod_{p \in S} \mathbb{Z}_p,$$

equipped with the action of $\mathbb{N}_S$ by multiplication.

Remark 5.1. This Y_S is the natural finite BC-type dynamical space. It is closer to the original BC construction than the adelic quotient X_S , because the latter is tailored to the explicit-formula and adèle-class-space picture, whereas the former is tailored to monoid dynamics and thermodynamic formalism.

5.2 Heuristic partition function

A BC-type monoid crossed product built from Y_S and $\mathbb{N}_S$ should have partition function

$$Z_S(\beta) = \sum_{n \in \mathbb{N}_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

This is exactly the finite Euler product from Section 3.

Remark 5.2 (Status). We do *not* construct the full monoid crossed-product C^* -algebra in this paper, nor do we classify its KMS states. Rather, we record Y_S as the correct finite BC-type dynamical candidate and emphasize that, at finite level, its expected partition function coincides with the rigorous arithmetic Euler product already proved in Section 3.

5.3 Archimedean factor

The finite BC-type space Y_S accounts naturally for the non-archimedean finite Euler product

$$\prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

It does *not* by itself account for the archimedean factor

$$\pi^{-\beta/2} \Gamma(\beta/2).$$

Thus if one wishes to recover the completed zeta factor at finite level, the real place must be incorporated separately. This separation is not a defect; it reflects the fact that the archimedean contribution is qualitatively different from the finite-prime BC thermodynamics.

Remark 5.3. This is one of the main reasons to distinguish X_S and Y_S . The arithmetic space X_S naturally remembers the real place through the finite S -adèle truncation, while the BC-type space Y_S naturally remembers the monoid of S -supported integers.

6 Bridge Between X_S and Y_S

We may now summarize the division of labor.

- The space X_S is the natural finite adelic/arithmetic configuration space. It is the correct stage for the finite explicit formula and for finite approximations to the adèle class space.
- The space Y_S is the natural finite BC-type dynamical configuration space. It is the correct stage for monoid thermodynamics, KMS heuristics, and finite Euler-product partition functions.

The common arithmetic object linking the two is the monoid

$$\Lambda_S = \mathbb{N}_S,$$

that is, the set of positive integers supported on primes in S .

From the point of view of X_S , this monoid appears through finite Euler products and truncated explicit-formula sums. From the point of view of Y_S , it appears as the monoid driving the dynamics.

Remark 6.1. The main open operator-algebraic task is to make this bridge precise. One would like to construct a genuine finite BC-type C^* -dynamical system based on Y_S , prove its KMS classification, and then show that its partition function and trace data reproduce the arithmetic identities proved on the X_S side.

7 Inductive Limit Picture and Hilbert–Pólya Motivation

As S grows, the spaces X_S approximate the full adèle class space, while the monoids Λ_S exhaust the positive integers:

$$\bigcup_S \Lambda_S = \mathbb{N}_{>0}.$$

Likewise,

$$\bigcup_S \mathbb{A}_{\mathbb{Q}}^{(S)}$$

is dense in $\mathbb{A}_{\mathbb{Q}}$, so the finite arithmetic spaces X_S form a natural approximation scheme to the full adèle-class-space picture.

At the level of arithmetic identities, the finite Euler products converge to the usual Euler product for $\zeta(\beta)$ for $\operatorname{Re}(\beta) > 1$, and the finite explicit formula converges formally to the full prime-power side as S increases.

Remark 7.1 (Heuristic Hilbert–Pólya interpretation). The finite- S systems should be regarded as computable shadows of a larger spectral picture. The point of the present note is not that a Hilbert–Pólya operator has been constructed, but that finite truncations already exhibit:

- Euler-product thermodynamics,
- von Mangoldt-weighted spectral sums,
- rapid numerical convergence,
- and a natural separation between arithmetic and dynamical configuration spaces.

These features suggest, but do not prove, that finite- S models may provide a useful approximation scheme for a future spectral framework.

Appendix A. Numerical Sandbox

The finite- S identities above are particularly well suited to computation. We record simple numerical routines illustrating the partial Euler products and the finite explicit formula for small S .

A.1 Partition function

For finite S , define

$$Z_S(\beta) = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

```
import mpmath as mp

def Z_S(beta, S):
    euler = mp.mpf('1')
    for p in S:
        euler *= 1 / (1 - mp.power(p, -beta))
    return euler

print(Z_S(2, [2,3,5]))
```

A.2 Prime-power side

For a Schwartz test function h , consider

$$\sum_{p \in S} \sum_{k \geq 1} \frac{\log p}{p^{k/2}} (h(k \log p) + h(-k \log p)).$$

```
import numpy as np

def h_np(t):
    return np.exp(-t**2)

def prime_side(S, num_terms=20):
    total = 0.0
    for p in S:
        logp = np.log(p)
        phalf = np.sqrt(p)
        pk = 1.0
        for k in range(1, num_terms + 1):
            pk *= phalf
            total += (logp / pk) * (h_np(k*logp) + h_np(-k*logp))
    return total

print(prime_side([2,3,5]))
```

A.3 Finite spectral/arithmetic side

This is the finite arithmetic side indexed by Λ_S :

$$\sum_{n \in \Lambda_S} \frac{\Lambda(n)}{\sqrt{n}} (h(\log n) + h(-\log n)).$$

```

def von_mangoldt(n: int) -> float:
    if n < 2:
        return 0.0
    p = 2
    if n % p == 0:
        m = n
        while m % p == 0:
            m //= p
        return np.log(p) if m == 1 else 0.0
    p = 3
    while p * p <= n:
        if n % p == 0:
            m = n
            while m % p == 0:
                m //= p
            return np.log(p) if m == 1 else 0.0
        p += 2
    return np.log(n)

def spectral_side(S, cutoff=500):
    Sset = set(S)
    total = 0.0
    for n in range(2, cutoff + 1):
        m = n
        ok = True
        p = 2
        if m % p == 0:
            if p not in Sset:
                ok = False
            while m % p == 0:
                m //= p
        p = 3
        while ok and p * p <= m:
            if m % p == 0:
                if p not in Sset:
                    ok = False
                while m % p == 0:
                    m //= p
            p += 2
        if not ok:
            continue
        if m > 1 and m not in Sset:
            continue
        vm = von_mangoldt(n)
        if vm != 0.0:
            ln = np.log(n)
            total += (vm / np.sqrt(n)) * (h_np(ln) + h_np(-ln))
    return total

print(spectral_side([2,3,5], cutoff=500))

```


A.4 Comment on the numerics

For small finite sets S , the prime-power side and the finite arithmetic side agree to high precision, as they must by Theorem 4.1. The numerical significance is not the identity itself, which is exact, but the speed with which the finite truncations stabilize as either the prime set S grows or the cutoff in n increases.

References

- [1] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1** (1995), 411–457.
- [2] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [3] A. Connes and M. Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*, American Mathematical Society, 2008.
- [4] H. M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974.
- [5] E. Bombieri, *The Riemann Hypothesis*, in *The Millennium Prize Problems*, Clay Mathematics Institute, 2000.
- [6] J. B. Conrey, *The Riemann Hypothesis*, Notices Amer. Math. Soc. **50** (2003), 341–353.
- [7] D. P. Williams, *Crossed Products of C^* -Algebras*, American Mathematical Society, 2007.
- [8] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.
- [9] M. Laca, S. Neshveyev, and M. Trifković, *Bost–Connes systems, Hecke algebras, and induction*, J. Noncommut. Geom. **7** (2013), 525–546.